

EROL AKARSU, Ph.D.

Lead Machine Learning Architect | Principal Software Engineer

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PROFESSIONAL SUMMARY

Lead Machine Learning Architect with a **Ph.D. in Computer Science (Distributed Computing)** and **25+ years** of enterprise experience delivering software, applications, and large-scale data platforms. Currently leading ML/AI initiatives at **Lowe's** — developing transformer-based image curation and Generative-AI project quota assistance — with a track record of **81+ production applications** across healthcare, legal tech, fintech, and e-commerce.

Specialize in designing and provisioning search infrastructure on Kubernetes and VM instances across **Google Cloud** and **AWS**. Built data pipelines for search data in **Apache Beam** and **Spark (Scala)**; architected distributed crawling (Apache Nutch) across 15K+ sites and a 1/2 TB indexed database (Apache SOLR) on Hadoop. Designed microservices in **Spring Boot**, **Node.js**, **Ruby on Rails**, and **JBoss**.

Recognized for enhancing cloud-based search with near-real-time indexing (Couchbase → Solr), leading cross-functional teams, and acting as main architect for moving search infrastructure to the cloud.

25+

Years Exp

81+

Apps Built

29+

AI Projects

9x

Perf Gain

Ph.D.

Distributed

TECHNICAL SKILLS

AI/ML:	LLMs, Transformers, GenAI, RAG, PyTorch, TensorFlow, NLP, Computer Vision, OpenAI API, Claude API, LangChain	Big Data:	Spark, Beam, Hadoop, Kafka, Scala, ETL
Search:	Elasticsearch, Solr, Nutch, Couchbase, Vector DBs	Cloud:	GCP, AWS, Terraform, Packer, Ansible, Jenkins
Backend:	Node.js, Express, Spring Boot, Python, Java, C++, C#	Frontend:	React, Next.js, TypeScript, Angular, Bootstrap
Mobile:	iOS/Swift, SwiftUI, StoreKit, Capacitor	Database:	PostgreSQL, Prisma, MongoDB, Redis, Oracle, MS-SQL
DevOps:	Docker, Docker Swarm, Kubernetes, CI/CD	Blockchain:	Web3, Solidity, Remix, NextJS

KEY ACHIEVEMENTS

- **9x performance gain** — reduced product indexing time from 360 to 40 minutes for 5M products using Spark/Scala ETL pipelines
- **Architected cloud migration** of enterprise search infrastructure serving Lowe's.com on Google Cloud
- **Built distributed 500GB+ search index** with Apache SOLR/Hadoop, crawling 15K+ sites
- **99.5% extraction success rate** migrating 700K lines of COBOL using XQUERY functional language
- **Created 81+ production AI/full-stack applications** across 15+ industries, demonstrating exceptional delivery velocity

PROFESSIONAL EXPERIENCE

Lowe's Companies, Inc.

Lead Machine Learning Architect

Charlotte, North Carolina

Jan 2024 - Present

- ▶ Lead development of ML transformer models for optimal image selection from a database of millions of product images, improving visual engagement across Lowes.com
- ▶ Architected Generative AI-powered Project Quota Assistance system enabling real-time Q&A for home improvement customers
- ▶ Designed and deployed LLM solutions processing thousands of daily user queries with high accuracy
- ▶ Collaborate with product, design, and engineering leadership to define AI/ML roadmap and strategy

Lowe's Companies, Inc.

Oct 2020 - Present

Principal Software Engineer

Mooresville, North Carolina

- ▶ Design and build features for Lowes.com including workflow applications for Apache Beam data pipelines and Elasticsearch frameworks
- ▶ Engineered pivotal e-commerce features (free delivery, expedited delivery, promotions) by leading end-to-end software lifecycle and developing Spark applications
- ▶ Spearheaded Spark application roadmap and launch plan, serving as single-threaded leader between senior VPs and engineering/design teams; prioritized end-user impact while maintaining regulatory compliance
- ▶ Significantly reduced client cost while elevating feature quality by planning and deploying automated build processes to optimize long-term maintenance
- ▶ Enhanced cloud-based search experience with near-real-time indexing; developed processes for ingesting products into Couchbase and indexing them on Solr
- ▶ Involved in project planning efforts including contracting, external contracting, and budget management

Wipro Limited

May 2018 - Oct 2020

Senior Software Architect

Richmond, Virginia

- ▶ Architected various products on Solr geared toward blockchain and cryptocurrency customers — profit-sharing with nominal interest fees, cutting-edge financing for Square lending, and security schema for borrowed tokens
- ▶ Utilized Solr, Couchbase, and Spark Scala to update product data, maintain data indices, and industrialize a 20+ step Spark Scala indexing pipeline
- ▶ Built Lowe's enterprise search infrastructure on Google Cloud using Terraform (resource creation), Packer (image creation), Ansible (provisioning), and Jenkins (auto-deployment)
- ▶ Established cloud-based load balancers, firewalls, and clusters while developing dozens of Jenkins jobs for Terraform scripts and Packer images
- ▶ **Reduced product indexing time 9x** (360 → 40 min for 5M products) by creating a Scala ETL process to transform product data across Couchbase, Spark, and Solr
- ▶ Enhanced operational efficiency with blockchain tooling: NextJS, Web3, Remix, and Solidity

pExchange

Dec 2017 - Mar 2018

Senior Software Developer

Richmond, Virginia

- ▶ Assessed and recommended a standard software toolset for the team, ensuring effective digital media workflow
- ▶ Designed a user management system for preparing presidential program budgets using Spring Boot, MongoDB, Angular, and Apache Shiro (roles & permissions)

Capital One

Aug 2016 - Dec 2017

Master Software Engineer

Richmond, Virginia

- ▶ Led design of data flows and sandbox environments for API development and service virtualization frameworks
- ▶ Orchestrated development of novel features including general simulators and Spring/Spring Boot frameworks while increasing end-user understanding of API networks
- ▶ Led team overhaul of developer.capitalone.com website and regression suite using Protractor, Cucumber, Spring, Node.js, Express, AngularJS, Bower/Gulp
- ▶ Produced fully automated API testing without user input via MVC, Swagger, FSM, Node.js, and Graph Computing

Self Employed — Independent Software Consultant

Full-Stack Developer / Architect (14 years)

- ▶ Oversaw many independent projects for high-priority customers including the **US Government**, applying industry web technologies and client product-management skills
- ▶ Renovated government website with Angular and Node.js — led full software development life cycle end to end with PMs and engineers
- ▶ Built an end-to-end Turkish search engine for a food retail chain: big-data pipelines, tokenizers, and 15 ranking criteria via Nutch / SOLR / Hadoop / Node.js / Bootstrap
- ▶ Developed SOA Router/Proxy in .NET (C#, LINQ, WCF, MS-SQL 2005, MS-IIS, Oracle 10g) to secure intra-company web services, with username/password, SSL certificates, and token authentication
- ▶ Supervised OpenDNS-based power-grid stimulation via a Python tool
- ▶ Integrated Java web services called from PL/SQL; designed new Navy portal with Oracle WCI / Web Center on Ubuntu Linux, UNIX, Windows
- ▶ Created Oracle WCI API for the Navy and automated portal deployments
- ▶ Delivered **99.5% extraction success rate** implementing rule extraction in XQUERY against 700K lines of COBOL (237 files)

Ferguson Enterprises

Jul 2007 - Jul 2016

Senior Software Analyst / Technical Lead (9 years)

- ▶ Led multiple enterprise web development projects across core business systems
- ▶ Served as technical lead, mentoring junior engineers and shaping delivery standards
- ▶ Partnered with business stakeholders to translate requirements into scalable solutions

LogicaCMG (formerly Cisco Systems)

Nov 1999 - Jul 2007

Senior Software Engineer / Lead Programmer (8 years)

Richmond, Virginia

- ▶ Led end-to-end development of Unified Messaging components — from Software Requirements Specification (SRS) and Software Design Specification (SDD) through release to the customer
- ▶ Architected and delivered a **highly-scalable unified messaging platform handling 300 simultaneous calls** using C++, XML, a specialized JSP-like template engine, and Apache Web Server on Linux/Solaris
- ▶ Authored IMAP wrapper and Telephony-API classes in **C++** with low-level TCP/IP socket communication; stack included Java SDK, PHP, JSP, XML, VoiceXML, Java LDAP, Java IMAP SDK, and JTAPI
- ▶ Designed and prototyped **Visual uOne** — a Model-Driven Architecture (MDA) environment letting users visually design voice applications in Microsoft Visio, translated to SVG/XML → XML/RDF/OWL documents in MySQL, with an engine generating VoiceXML statically or dynamically at runtime
- ▶ Architected MVC framework for customized voice/video front-ends: VoiceXML pages generated dynamically from LDAP user preferences and IMAP messages via an Apache C++ back-end plug-in driven by abstract XML application definitions
- ▶ Diagnosed and resolved a production memory leak / corruption issue in uOne by modifying Apache's *worker* MPM to synchronize and recycle worker threads after a tuned number of voice calls
- ▶ Built an automated customer release-document generator using Ruby, XQUERY, XML, and the IBM ClearCase API — fetching defect data, gathering per-server package info, and populating release templates
- ▶ Implemented content management for the product and services catalog using Java/RDF/XML/XQUERY/OWL web-service definitions backed by LDAP and SQL
- ▶ Triaged incoming customer defects, change requests, and new-feature requests: analyzed scope, estimated cost, and delivered solutions on aggressive timelines
- ▶ Gathered and reconciled conflicting customer requirements; served as technical lead for CMG Wireless's Unified Communications and Unified Messaging software across analysis, design, development, and testing

SELECTED AI / FULL-STACK PROJECTS

AI Churn Prediction System
Customer Analytics

AI Insurance Agency Platform
InsurTech

- ▶ SaaS platform predicting customer churn with AI risk scores, segmentation, and intervention recommendations (React, Node.js, PostgreSQL, Claude AI)

AI Due Diligence Platform M&A / Investment

- ▶ M&A analysis with document parsing, risk assessment, and automated reporting for investment decisions (React, Node.js, OpenRouter, NLP)

AI Inventory Forecasting Supply Chain

- ▶ Demand forecasting with supplier scoring, restock recommendations, and anomaly detection (React, Express, Claude AI, PostgreSQL)

AI Regulatory Compliance Compliance

- ▶ Regulation tracking, audit preparation, risk assessment, and automated compliance reporting (React, Node.js, Claude AI)

- ▶ Full agency management with AI claims assistant, coverage analyzer, risk assessor, and voice receptionist (Next.js, TypeScript, Prisma, PostgreSQL)

AI Contract Analysis Legal Tech

- ▶ Clause extraction, risk identification, negotiation suggestions, and term comparison across documents (React, Node.js, NLP, Claude AI)

AI Voice Agent Conversational AI

- ▶ Voice-enabled AI assistant with speech recognition, natural language understanding, and response generation (Twilio, OpenAI, React, Node.js)

AI Recruiter Screener HR Tech

- ▶ Resume parsing, skill matching, interview question generation, and bias-free evaluation scoring (React, Node.js, NLP)

AI Application Portfolio — 81+ Production Applications, 15+ Industries

Business & Finance AI: Churn Prediction, Pricing Optimizer, Financial Reports, Due Diligence, Invoice Processing, Tax Prep

Document & Content AI: Contract Negotiation, Proposal/SOW Generator, Grant Writer, Knowledge Base Wiki, Code Documentation

Marketing & Sales AI: Content Studio, Marketing Automation, Sales Outreach, Competitive Analysis, Social Proof Generator

Visual & Media AI: Menu Digitizer, Floor Plan Analyzer, Product Photo Enhancer, Video Testimonial Creator, Visual QA Inspector

Operations & Support AI: Customer Support Agent, Email Triage, Review Response, Inventory Forecasting, Appointment Scheduler

Industry Solutions: Healthcare, Real Estate, Legal Forms, E-Commerce, Food Service, Beauty & Wellness, Gov Contracts

Tech stack: React, Next.js, Express, Node.js, Swift, PostgreSQL, Prisma, Stripe, Docker, OpenAI API, Claude API. Full portfolio of 50+ production-ready AI applications available upon request.

EDUCATION

Ph.D., Computer Science — IDE for Grid / Distributed Computing

Syracuse University, NY · 1994 - 1999 · Thesis implemented in C++

M.S., Computer Science

Syracuse University, NY · 1994 - 1996

M.S., Computer Engineering

Istanbul Technical University · 1991 - 1993

B.S., Computer Engineering

Ege University · 1986 - 1991

CERTIFICATIONS & TOP SKILLS

- Neural Networks and Deep Learning
- Machine Learning: Regression and Classification
- Machine Learning: Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning

Agile Methodologies

Software Engineering

Software Development

DevOps Management

Big-Data Architecture

Cross-Functional Collaboration

- Stanford/Coursera ML Specialization (2022)

User Research

Agile Delivery

PUBLICATIONS

PhD Thesis & complete list of publications available on LinkedIn profile.

GitHub: github.com/eakarsu | **LinkedIn:** linkedin.com/in/erol-akarsu